

The simulation is governed by 2 systems:

- The inversion of the inactive (R)-enantiomer
- The removal of the (S)-enantiomer

- Let $R(t)$ represent mass (mg) of (R)-enantiomer remaining in body after t (hours)
- Let $S(t)$ represent mass (mg) of (S)-enantiomer remaining in body after t (hours)
- Let k_{inv} be constant rate (hr^{-1}) for inversion of (R)-enantiomer into (S)-enantiomer in the liver
- Let k_{el} be constant rate (hr^{-1}) for removal of (S)-enantiomer via renal clearance

Rate of depletion of (R)-enantiomer is:

$$\frac{dR}{dt} = (-k_{inv}) R(t)$$

Rate of generation of (S)-enantiomer is:

$$\frac{dS}{dt} = R(t)(k_{inv}) - S(t)(k_{el})$$

Rate = generation - removal

Differential Equations

$$\frac{dR}{dt} = -k_{inv} (R(t))$$

$$\frac{1}{R(t)} dR = -k_{inv} dt$$

$$\ln |R(t)| = -k_{inv} t + C$$

$$R(t) = e^{C - k_{inv} t} \quad \text{let } e^C = A$$

$$\boxed{R(t) = A e^{-k_{inv} t}}$$

where A = initial mass of (R)-enantiomer

$$\frac{dS}{dt} = R(t)(k_{inv}) - S(t)(k_{el})$$

$$= (A e^{-k_{inv} t})(k_{inv}) - S(t)(k_{el})$$

$$\frac{dS}{dt} + S(t)(k_{el}) = k_{inv}(A e^{-k_{inv} t})$$

$$\rightarrow \frac{dS}{dt} + P(t)y = Q(t)$$

$$\begin{aligned} \text{Integral factor} &= e^{\int P(t) dt} \\ &= e^{\int k_{el} dt} \\ &= e^{k_{el} t} \end{aligned}$$

$$e^{k_{e1}t} \left(\frac{dS}{dt} \right) + e^{k_{e1}t} (k_{e1} S(t)) = e^{k_{e1}t} (k_{inv} (Ae^{-k_{inv}t}))$$

$$e^{k_{e1}t} \frac{dS}{dt} + k_{e1} e^{k_{e1}t} S(t) = k_{inv} A e^{(k_{e1} - k_{inv})t}$$

$$\begin{aligned} u &= S(t) \\ v &= e^{k_{e1}t} \end{aligned} \quad \frac{dv}{dt} = k_{e1} e^{k_{e1}t}$$

$$\frac{du}{dt} v + \frac{dv}{dt} u = \text{Product rule} = \frac{d}{dt} (u \times v)$$

$$\therefore e^{k_{e1}t} \frac{dS}{dt} + k_{e1} e^{k_{e1}t} S(t) = \frac{d}{dt} (S(t) e^{k_{e1}t})$$

$$\frac{d}{dt} (S(t) e^{k_{e1}t}) = k_{inv} A e^{(k_{e1} - k_{inv})t}$$

$$S(t) e^{k_{e1}t} = \int k_{inv} A e^{(k_{e1} - k_{inv})t} dt$$

$$S(t) e^{k_{e1}t} = \frac{k_{inv} A}{k_{e1} - k_{inv}} e^{(k_{e1} - k_{inv})t} + C$$

$$S(t) = e^{-k_{e1}t} \left(\frac{k_{inv} A}{k_{e1} - k_{inv}} e^{(k_{e1} - k_{inv})t} + C \right)$$

$$S(t) = \frac{k_{inv} A}{k_{e1} - k_{inv}} e^{-k_{inv}t} + C e^{-k_{e1}t}$$

$$S(t) = \frac{k_{inv} A}{k_{e1} - k_{inv}} e^{-k_{inv} t} + c e^{-k_{e1} t} \quad \text{let } t = 0$$

$$S_0 - \frac{k_{inv} A}{k_{e1} - k_{inv}} = c$$

$$S(t) = \left(\frac{k_{inv} A}{k_{e1} - k_{inv}} \right) e^{-k_{inv} t} + \left(S_0 - \frac{k_{inv} A}{k_{e1} - k_{inv}} \right) e^{-k_{e1} t}$$

To simulate you need Euler's Method

Euler's Method:

$$\text{New} = \text{Old} + (\text{Rate of change} \times \text{time})$$

$$R(t + \Delta t) = R(t) + (-k_{\text{inv}} R(t)) \Delta t$$

$$\text{New mass} = \text{Old mass} - \text{Rate of change} \times \text{time}$$

→ This is equation of depletion of (R)-enantiomer

$$S(t + \Delta t) = S(t) + (k_{\text{inv}} R(t) - k_{\text{el}} S(t)) \Delta t$$

$$\text{New mass} = \text{Old mass} + (\text{generation of (S)} - \text{usage of (S)}) \times \text{time}$$

→ This is equation of depletion of (S)-enantiomer